

# cmd\_and\_clients

## CLI / TUI / Web / Desktop / Mobile clients + cmd tools

Inventory slice for `helix_code/cmd/*` (cmd tools) and `helix_code/applications/*` + `helix_code/web/*` (client applications). Assessed from source evidence per the anti-bluff covenant (CONST-035 / §11.4.107). 📺 **Video is no for every row** — device/display recordings are only just starting; the conductor owns video confirmation. Overall is therefore never confirmed; honest rollups are working-untaped / partial / gap.

Recordability per client (for the conductor): **CLI, TUI, Web = feasible now** (terminal + headless HTTP). **Desktop (Fyne) = host-display required**. **Mobile: Aurora OS / Harmony OS** are Go/Fyne desktop-class binaries (recordable on a Linux display); **Android / aurora\_os HAP / harmony\_os HAP need device/emulator; iOS needs a built app (no Xcode project present → not buildable yet)**.

| Area             | Component    | Feature                                                                                             | Dev  | Wired | Real- |
|------------------|--------------|-----------------------------------------------------------------------------------------------------|------|-------|-------|
| application(cli) | cmd/cli REPL | Plain-text prompt → real LLM (streaming via <code>provider.GenerateStream</code> )                  | done | yes   | yes   |
| application(cli) | cmd/cli REPL | <code>/generate</code> real LLM generation (BLUFF-001 resolved)                                     | done | yes   | yes   |
| application(cli) | cmd/cli REPL | <code>/models</code> list models via <code>providerManager.GetProviders</code> (BLUFF-002 resolved) | done | yes   | yes   |
| application(cli) | cmd/cli REPL | <code>/workers</code> list workers                                                                  | done | yes   | yes   |
| application(cli) | cmd/cli REPL | <code>/health</code> health check                                                                   | done | yes   | yes   |
| application(cli) | cmd/cli REPL | <code>/diff [ref]</code> real git diff (os/exec)                                                    | done | yes   | yes   |
| application(cli) | cmd/cli REPL | <code>/debate &lt;prompt&gt;</code> DebateOrchestrator over real provider                           | done | yes   | yes   |
| application(cli) | cmd/cli REPL | <code>/specify &lt;request&gt;</code> HelixSpecifier Specify phase over real provider               | done | yes   | yes   |
| application(cli) | cmd/cli REPL | <code>/checkpoint [create/list/restore]</code> workspace snapshots                                  | done | yes   | yes   |
| application(cli) | cmd/cli REPL | <code>/undo</code> revert last action                                                               | done | yes   | yes   |
| application(cli) | cmd/cli REPL |                                                                                                     | done | yes   | yes   |

| Area             | Component                  | Feature                                                            | Dev  | Wired | Real |
|------------------|----------------------------|--------------------------------------------------------------------|------|-------|------|
|                  |                            | <code>/clear /reset</code> clear conversation history              |      |       |      |
| application(cli) | cmd/cli REPL               | <code>/help help; /exit /quit</code> exit                          | done | yes   | yes  |
| application(cli) | cmd/cli REPL               | @-file mentions (real os.ReadFile context attach)                  | done | yes   | yes  |
| application(cli) | cmd/cli REPL               | Context-window tracking + generation stats                         | done | yes   | yes  |
| application(cli) | cmd/cli commands           | <code>list / show / run / reload</code> markdown commands          | done | yes   | yes  |
| application(cli) | cmd/cli hooks              | <code>list / enable / disable / validate / test</code> git hooks   | done | yes   | yes  |
| application(cli) | cmd/cli lsp                | <code>status / list-servers / restart / stop</code> LSP servers    | done | yes   | yes  |
| application(cli) | cmd/cli mcp                | <code>add / remove / list / test / auth / logs</code> MCP servers  | done | yes   | yes  |
| application(cli) | cmd/cli permissions        | <code>list / add / remove / check</code> tool permissions          | done | yes   | yes  |
| application(cli) | cmd/cli sessions           | <code>list / show / delete</code> sessions                         | done | yes   | yes  |
| application(cli) | cmd/cli skills             | <code>list / show / invoke / reload</code> skills                  | done | yes   | yes  |
| service          | cmd/server                 | HTTP/gRPC/WebSocket server boot (bin/helixcode)                    | done | yes   | yes  |
| application(web) | web/frontend               | LLM generate console (POST <code>/api/v1/llm/generate</code> )     | done | yes   | yes  |
| application(web) | web/frontend               | LLM streaming console (SSE, POST <code>/api/v1/llm/stream</code> ) | done | yes   | yes  |
| application(web) | web/frontend               | Specify phase form (POST <code>/api/v1/specify</code> )            | done | yes   | yes  |
| application(web) | web/frontend               | Response/metadata rendering (no client simulation)                 | done | yes   | yes  |
| service          | internal/server (HTTP API) | <code>/api/v1/llm/generate</code> real provider.Generate           | done | yes   | yes  |
| service          | internal/server (HTTP API) | <code>/api/v1/llm/stream</code> real provider.GenerateStream (SSE) | done | yes   | yes  |
| service          |                            |                                                                    | done | yes   | yes  |

| Area             | Component                  | Feature                                                                                   | Dev  | Wired | Real |
|------------------|----------------------------|-------------------------------------------------------------------------------------------|------|-------|------|
|                  | internal/server (HTTP API) | /api/v1/specify real speckit<br>Specify phase                                             |      |       |      |
| service          | internal/server (HTTP API) | /api/v1/llm/providers, /llm/<br>models list                                               | done | yes   | yes  |
| service          | internal/server (HTTP API) | /api/v1/auth register / login /<br>logout / refresh (JWT)                                 | done | yes   | yes  |
| service          | internal/server (HTTP API) | /api/v1/users me get/update/<br>delete                                                    | done | yes   | yes  |
| service          | internal/server (HTTP API) | /api/v1/workers CRUD +<br>heartbeat + metrics                                             | done | yes   | yes  |
| service          | internal/server (HTTP API) | /api/v1/tasks CRUD + assign/<br>start/complete/fail/retry/<br>checkpoint                  | done | yes   | yes  |
| service          | internal/server (HTTP API) | /api/v1/projects CRUD +<br>sessions + planning/building/<br>testing/refactoring workflows | done | yes   | yes  |
| service          | internal/server (HTTP API) | /api/v1/sessions CRUD                                                                     | done | yes   | yes  |
| service          | internal/server (HTTP API) | /api/v1/system stats + status                                                             | done | yes   | yes  |
| service          | internal/server (HTTP API) | /api/v1/memory systems +<br>stats                                                         | done | yes   | yes  |
| service          | internal/server (HTTP API) | /api/v1/qa session start/list/<br>status/report/screenshot/cancel                         | done | yes   | unkn |
| service          | internal/server (HTTP API) | /api/v1/screenshot engines +<br>capture                                                   | done | yes   | unkn |
| service          | internal/server (HTTP API) | /ws MCP WebSocket bridge                                                                  | done | yes   | yes  |
| service          | internal/server (HTTP API) | /health, /api/v1/health, /<br>metrics, /api/v1/server/info                                | done | yes   | yes  |
| service          | internal/server (HTTP API) | /debug/pprof/* (opt-in<br>profiling)                                                      | done | yes   | yes  |
| application(tui) | applications/terminal_ui   | LLM chat (real provider,<br>verifier-driven model<br>discovery)                           | done | yes   | yes  |
| application(tui) | applications/terminal_ui   | Dashboard / Tasks / Workers /<br>Projects / Sessions panels (real<br>managers)            | done | yes   | yes  |
| application(tui) | applications/terminal_ui   | Sidebar nav + key bindings (d/<br>t/w/p/s/l/q/c)                                          | done | yes   | yes  |
| application(tui) | applications/terminal_ui   |                                                                                           | done | yes   | yes  |

| Area                 | Component                         | Feature                                                                   | Dev     | Wired   | Real    |
|----------------------|-----------------------------------|---------------------------------------------------------------------------|---------|---------|---------|
|                      |                                   | Skill dispatcher + tool registry (git/fs/grep/LSP/MCP, graceful-nil)      |         |         |         |
| application(tui)     | applications/terminal_ui          | HelixMemory durable cross-session store (SQLite, default-on)              | done    | yes     | yes     |
| application(tui)     | applications/terminal_ui          | Status bar (DB/Redis/LLM status, context %), notifications, themes, i18n  | done    | yes     | yes     |
| application(tui)     | applications/terminal_ui          | QA engine panel (helixqa.Engine wired)                                    | partial | yes     | unknown |
| application(desktop) | applications/desktop (Fyne)       | LLM chat tab (real provider, verifier-driven models)                      | done    | yes     | yes     |
| application(desktop) | applications/desktop (Fyne)       | Dashboard / Tasks / Workers / Projects / Sessions tabs (real managers)    | done    | yes     | yes     |
| application(desktop) | applications/desktop (Fyne)       | Settings tab (theme, server config, shortcuts; desktop.yaml)              | done    | yes     | yes     |
| application(desktop) | applications/desktop (Fyne)       | Agentic tools + skills/plugins wiring (graceful-nil)                      | done    | yes     | yes     |
| application(desktop) | applications/desktop (Fyne)       | NoGUI build mode (-tags nogui CLI fallback)                               | done    | yes     | yes     |
| application(desktop) | applications/desktop (Fyne)       | TUI-parity wiring (parity_wiring_test.go)                                 | done    | yes     | yes     |
| application(mobile)  | applications/aurora_os (Go/Fyne)  | Full multi-page GUI: dashboard/projects/sessions/tasks/LLM/workers/system | done    | yes     | yes     |
| application(mobile)  | applications/aurora_os (Go/Fyne)  | NoGUI CLI mode (cobra) + theme system                                     | done    | yes     | yes     |
| application(mobile)  | applications/harmony_os (Go/Fyne) | 10-tab GUI incl. distributed engine + multi-device scheduling             | done    | yes     | yes     |
| application(mobile)  | applications/harmony_os (Go/Fyne) | NoGUI CLI + interactive shell + theme system                              | done    | yes     | yes     |
| application(mobile)  | applications/android (Kotlin)     | Connect + task list (RecyclerView) over Go mobile-core bridge             | partial | partial | no      |
| application(mobile)  | applications/android (Kotlin)     | Models / settings / notifications / theme UI                              | stub    | no      | no      |
| application(mobile)  | applications/ios (Swift)          | Connect + task list (UITableView) over Go mobile-core bridge              | partial | partial | no      |

| Area                | Component                    | Feature                                                    | Dev  | Wired | Real |
|---------------------|------------------------------|------------------------------------------------------------|------|-------|------|
| application(mobile) | applications/ios (Swift)     | Models / settings / notifications / theme UI               | stub | no    | no   |
| application(cli)    | cmd/helix_config             | Interactive provider/credential config wizard → YAML/.env  | done | yes   | yes  |
| application(cli)    | cmd/config_test              | Config + provider-credential validator                     | done | yes   | yes  |
| application(cli)    | cmd/i18n                     | i18n bundle/translator tooling (library, no main)          | done | yes   | yes  |
| application(cli)    | cmd/infrastructure           | Container / k8s / registry readiness checks                | done | yes   | unkn |
| application(cli)    | cmd/performance_optimization | pprof profiling + bottleneck analysis + benchmarks         | done | yes   | yes  |
| application(cli)    | cmd/security_test            | Security test harness (PoC execution + verification)       | done | yes   | yes  |
| application(cli)    | cmd/security_fix             | Finding ingestion + policy-driven fix + re-scan validation | done | yes   | yes  |
| application(cli)    | cmd/security_fix_standalone  | Batch parallel security fix + audit-trail reporting        | done | yes   | yes  |
| application(cli)    | cmd/security_scan            | AST/pattern code scan + leak detect + SARIF report         | done | yes   | unkn |

## Honesty notes (anti-bluff)

- **cmd/security\_scan has ZERO tests** (only `main.go`) — marked `Tests=none`, `Overall=gap`. Untested scanner = bluff risk per CONST-048/CONST-050.
- **iOS / Android are scaffolds**: real source (JSON parse, list bind, Go mobile-core bridge) but single-screen, hardcoded localhost test server, **no Xcode project / no Gradle+manifest** → not buildable, not recordable. `Real-use=no`, `Overall=gap`.
- **Aurora OS / Harmony OS are genuine Go/Fyne apps** (buildable via `make aurora-os / make harmony-os`, comprehensive unit+integration tests) — but integration features need real PostgreSQL/Redis/LLM backends, and full HAP/multi-device exercise needs the actual OS environment.
- **Web LLM endpoints** (`/llm/generate`, `/llm/stream`, `/specify`) have real Ollama-backed e2e tests (`tests/integration/{llm_generate, llm_stream, specify_server}_e2e_test.go`, build tag `integration`, honest SKIP-OK when Ollama unreachable) → `Tests=e2e`, `V&V=yes` (`docs/qa/web-llm-e2e-20260615/`). The 60+ CRUD/auth/workflow endpoints are real but tested at the manager/service layer, not at the HTTP-transport layer → `Tests=integ/none` honestly.
- **No row is confirmed** — every 📺 `Video=no`; rollups are working-untaped (real + tested, no video), `partial` (real but thin/unverified test coverage), or `gap` (scaffold or untested).

**Feature count: 67 rows** (CLI REPL 14, CLI subcommand groups 7, server boot 1, web frontend 4, HTTP API groups 18, TUI 7, desktop 6, mobile 8, other cmd tools 11 + security\_scan 1).